LECTURE 7

STATISTICAL MECHANICS FOR STATISTICIANS

Saifuddin Syed

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- Today we will see it from the perspective of statistical mechanics, where these ideas originated

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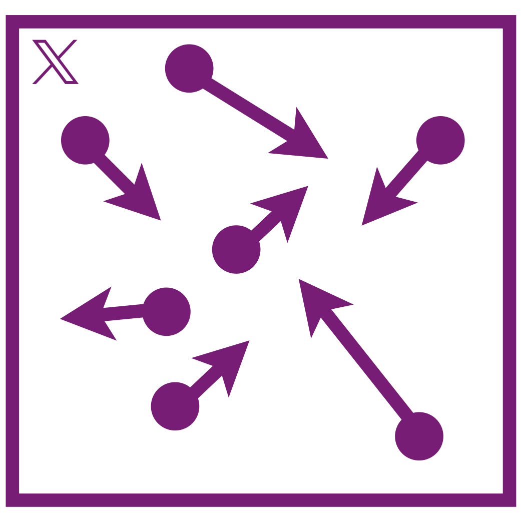
  - A particle i has mass m with position q_i and momentum p_i



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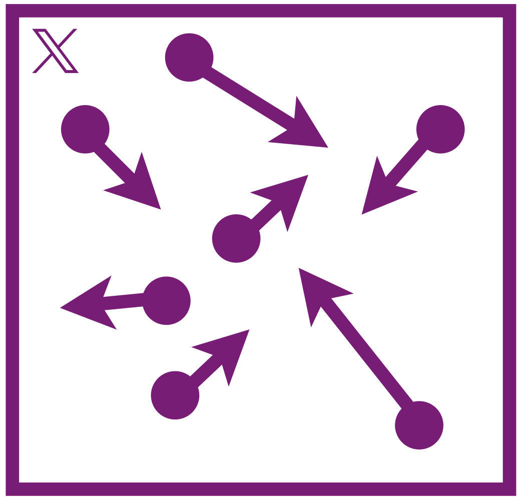
- The system will evolve accoring to Hamiltonion dynamics, $x = (q_{1:N}, p_{1:N}) \in \mathbb{R}^{6N} = \mathbb{X}$

$$H(x) = \sum_i V(q_i) + \frac{1}{2m} \sum_i |p_i|^2$$



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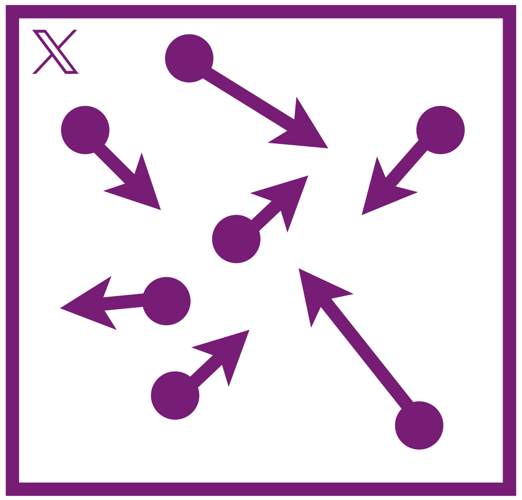

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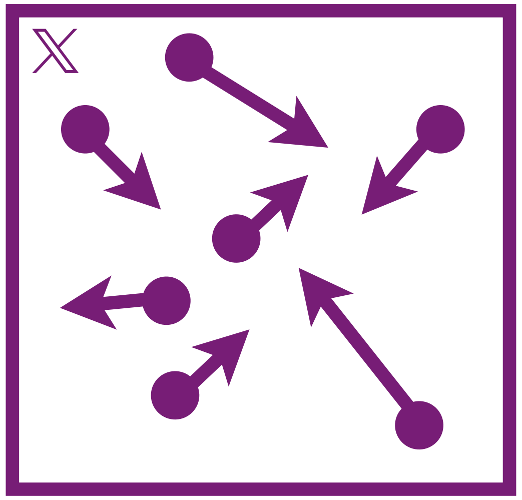
  - The stationary distribution of the particles satisfies :

$$\eta(x) \propto \exp(-H(x))$$



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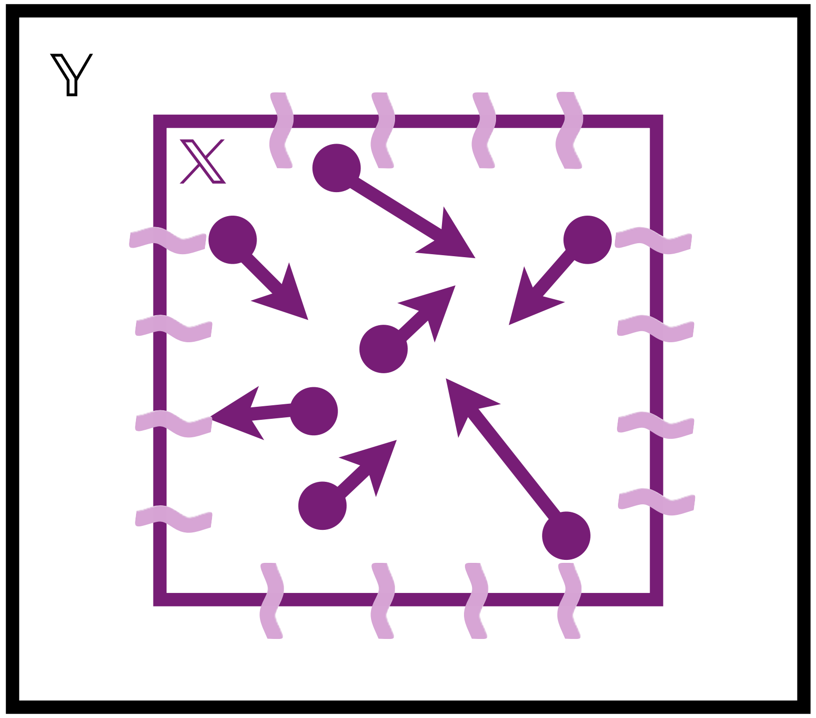

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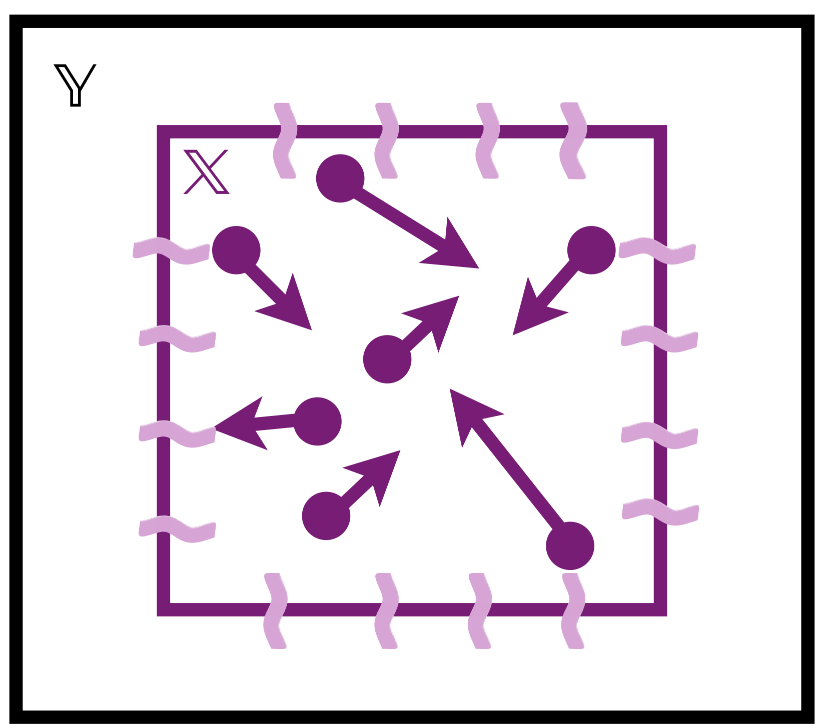

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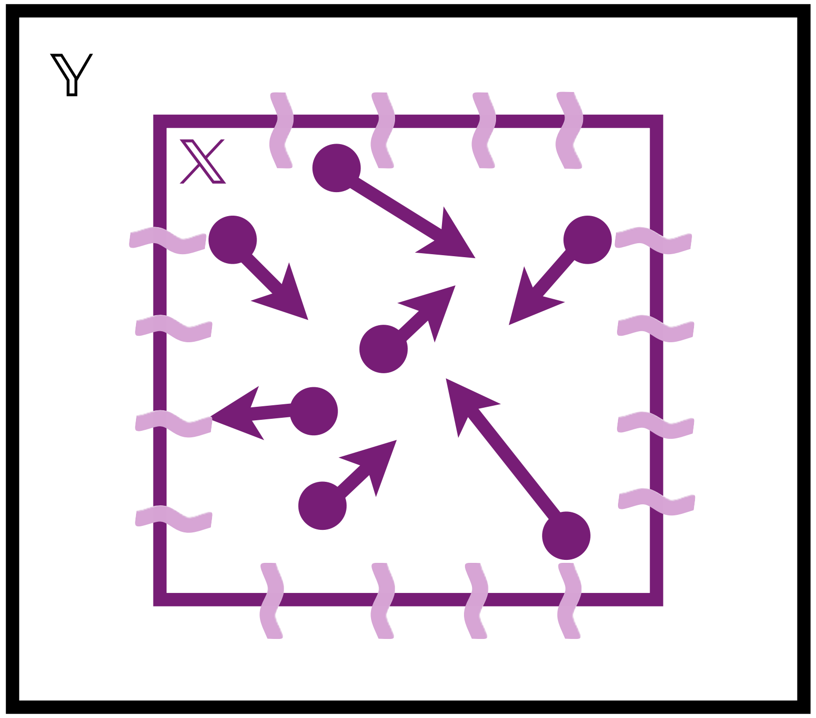

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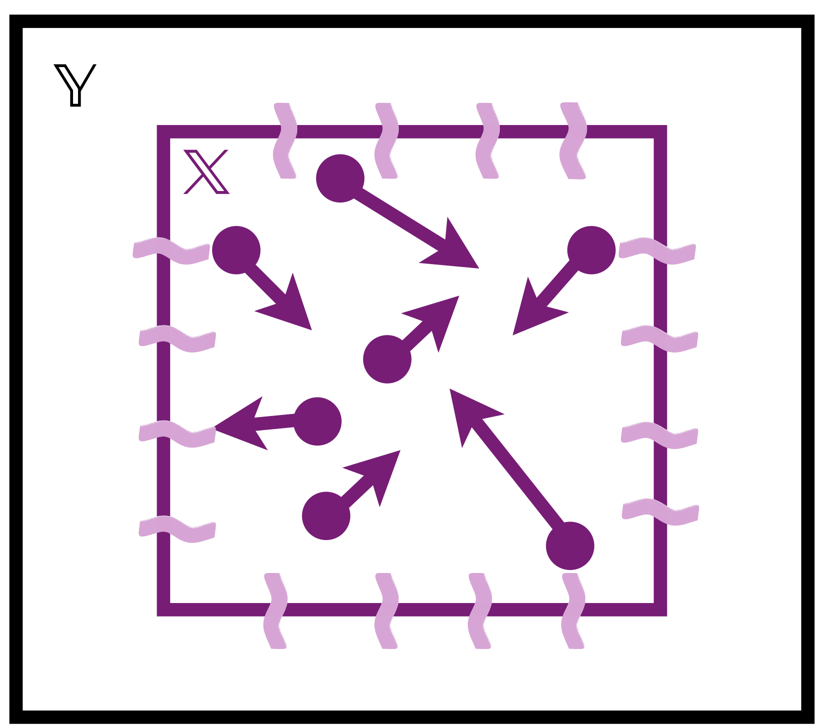

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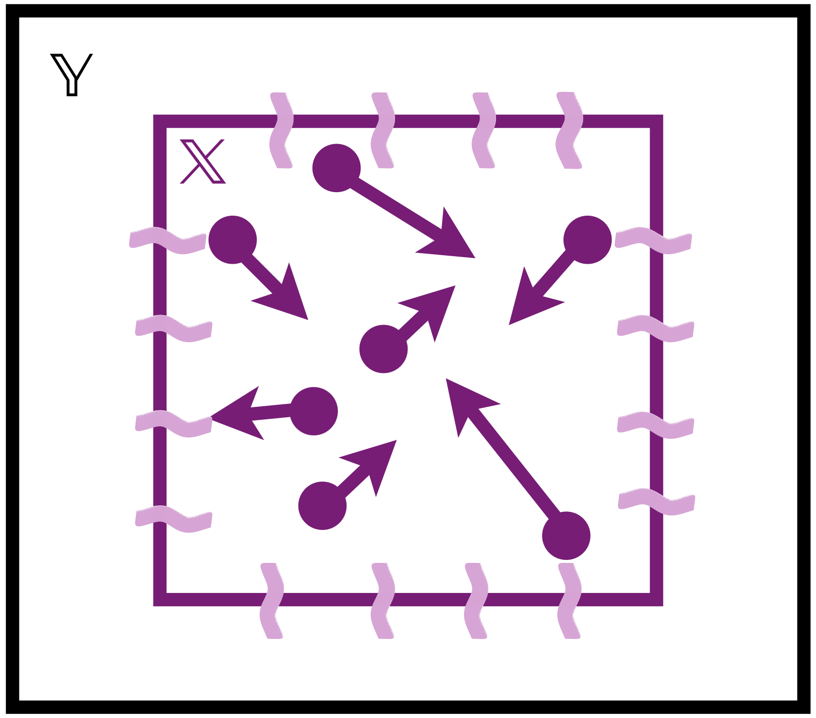
- We can't observe the resevoir states $y \in \mathbb{Y}$, but can measure the energy added or taken away X
- What is the law π of the system $x \in \mathbb{X}$ after the system reaches equilibrium?

$$\pi(x) = \eta(x \,|\, y)$$



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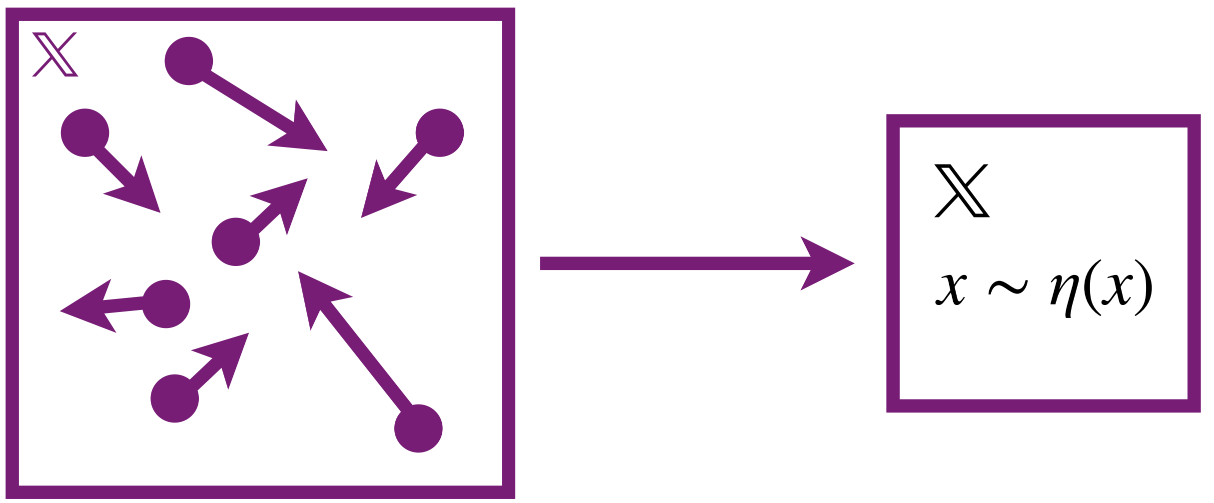

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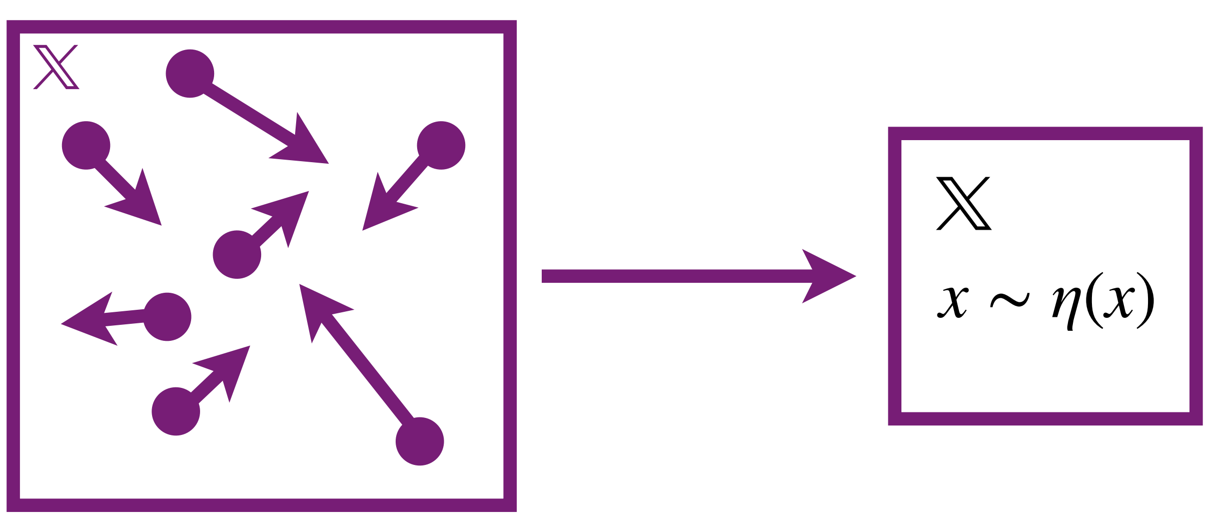

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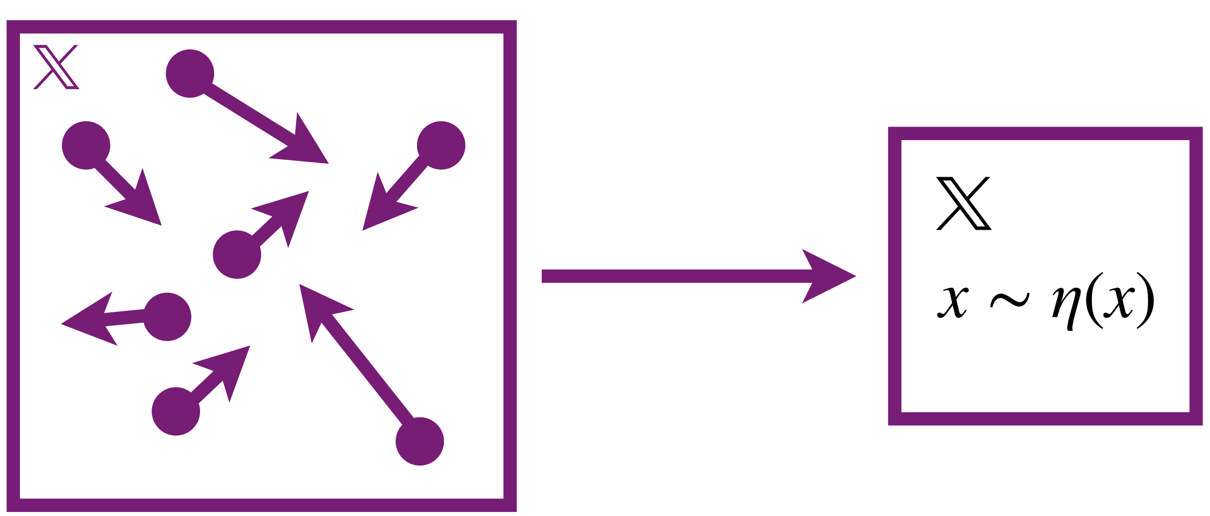

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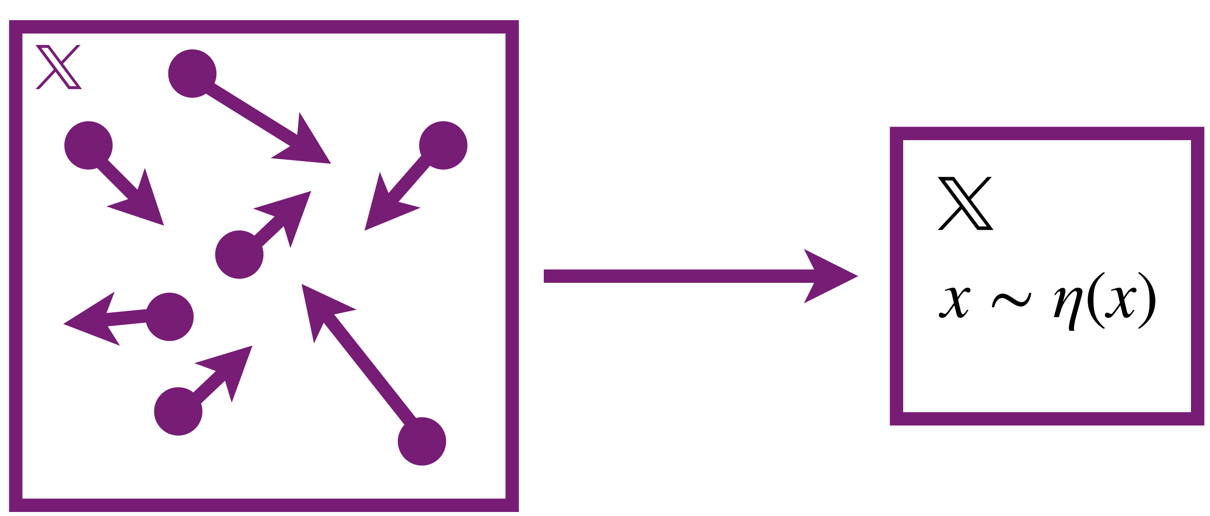

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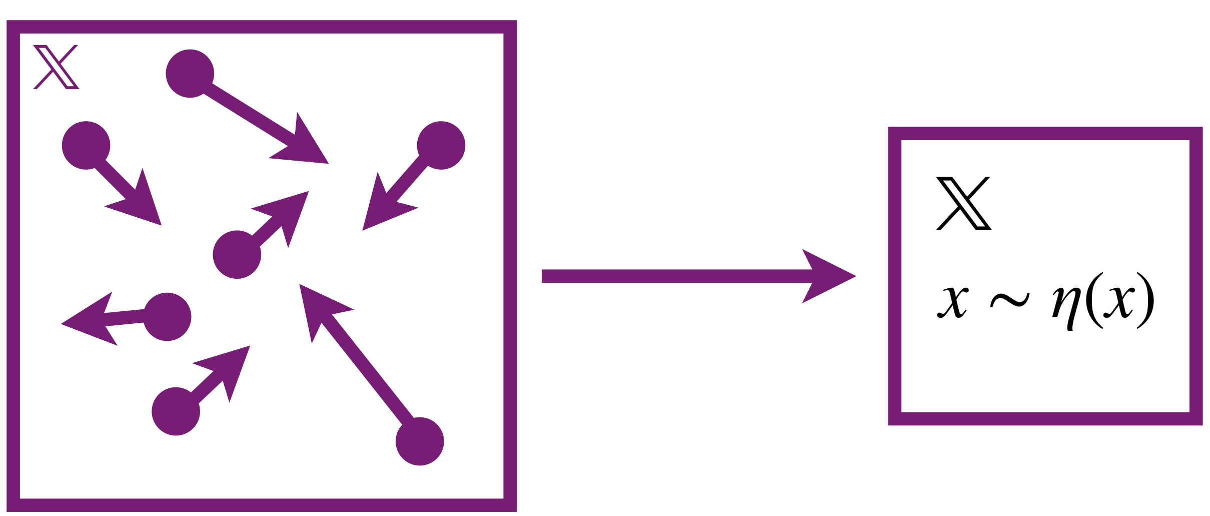

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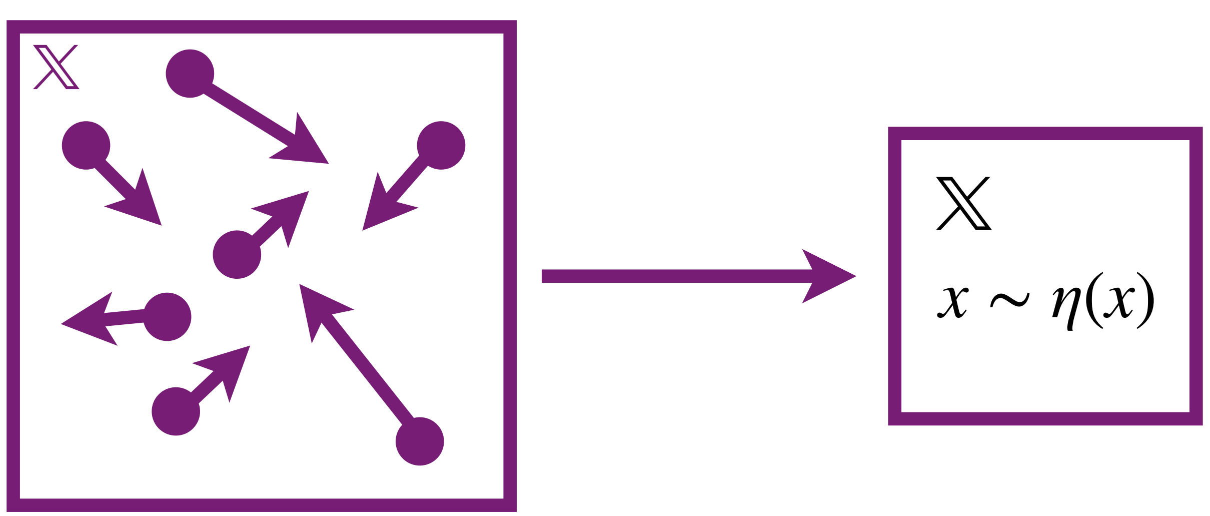
  - **Example:** the state of your molecule before a reaction occurs



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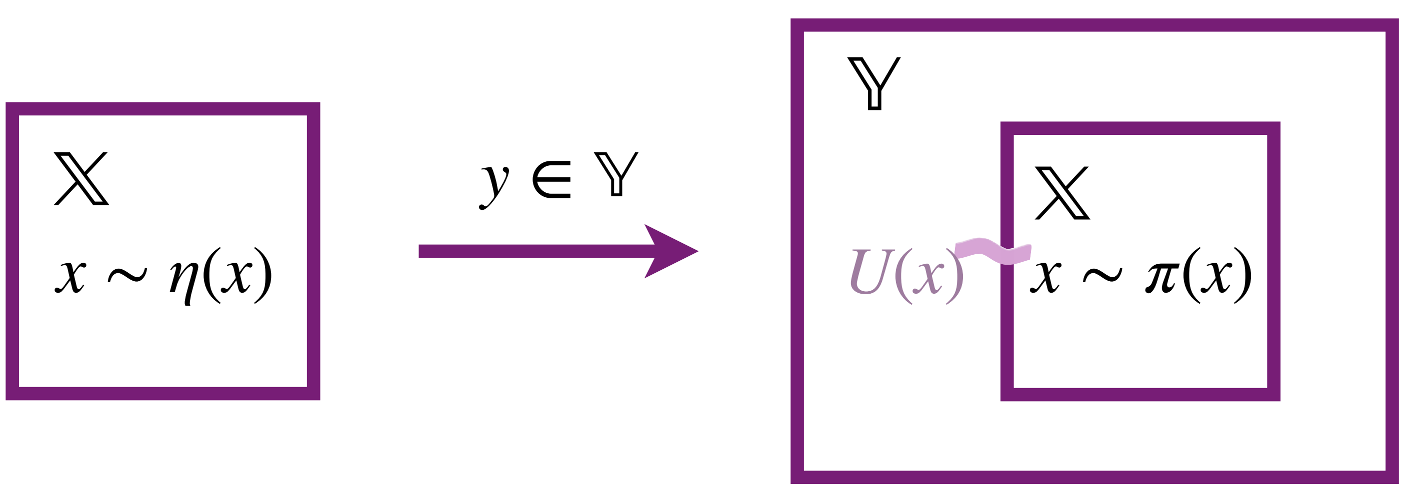

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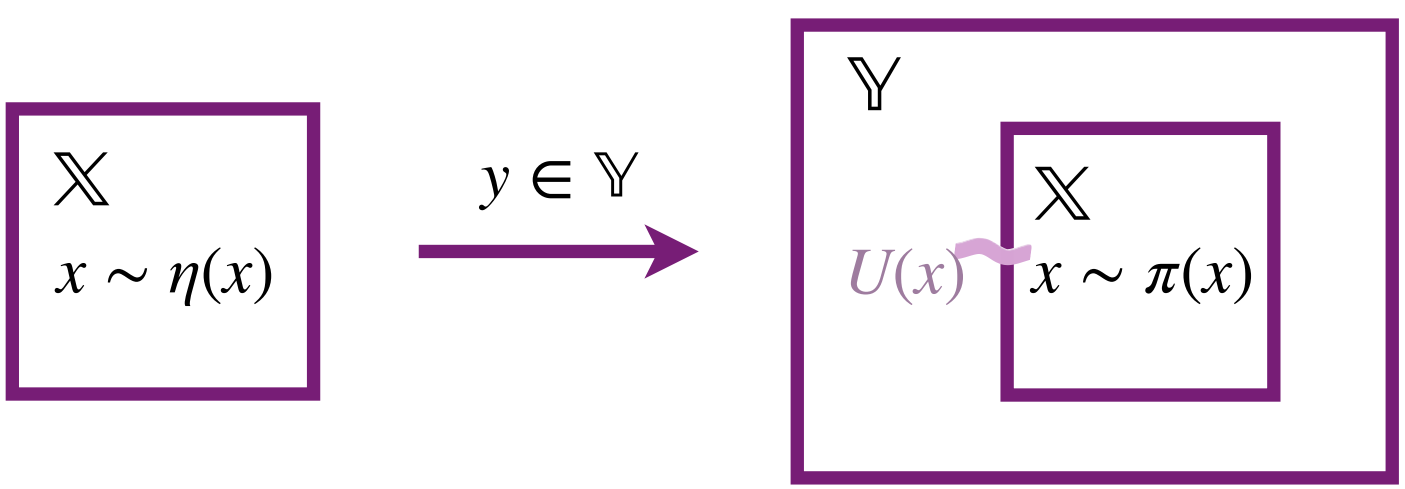

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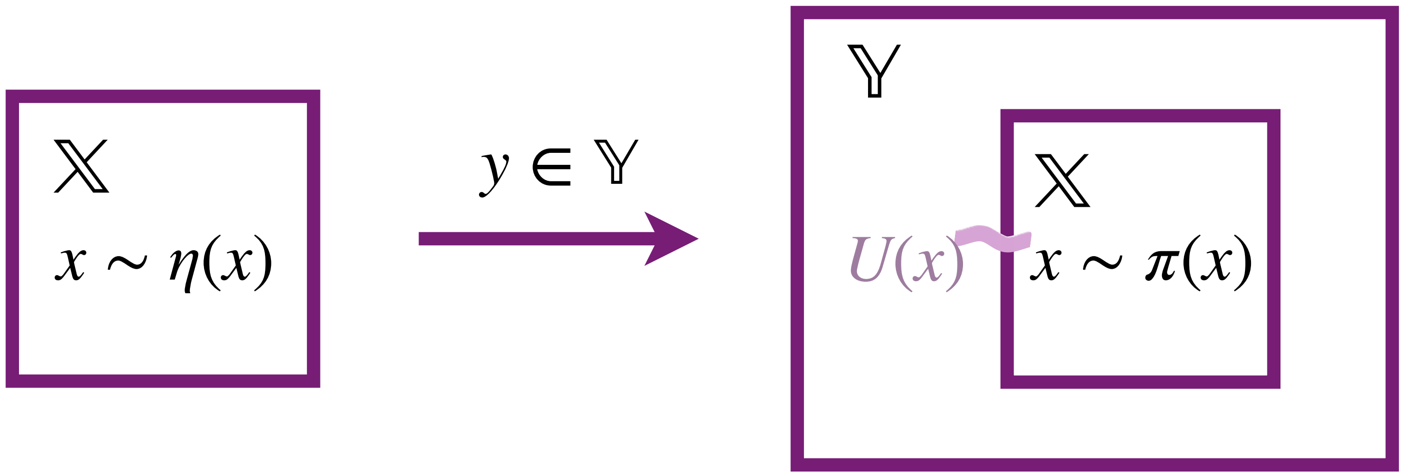

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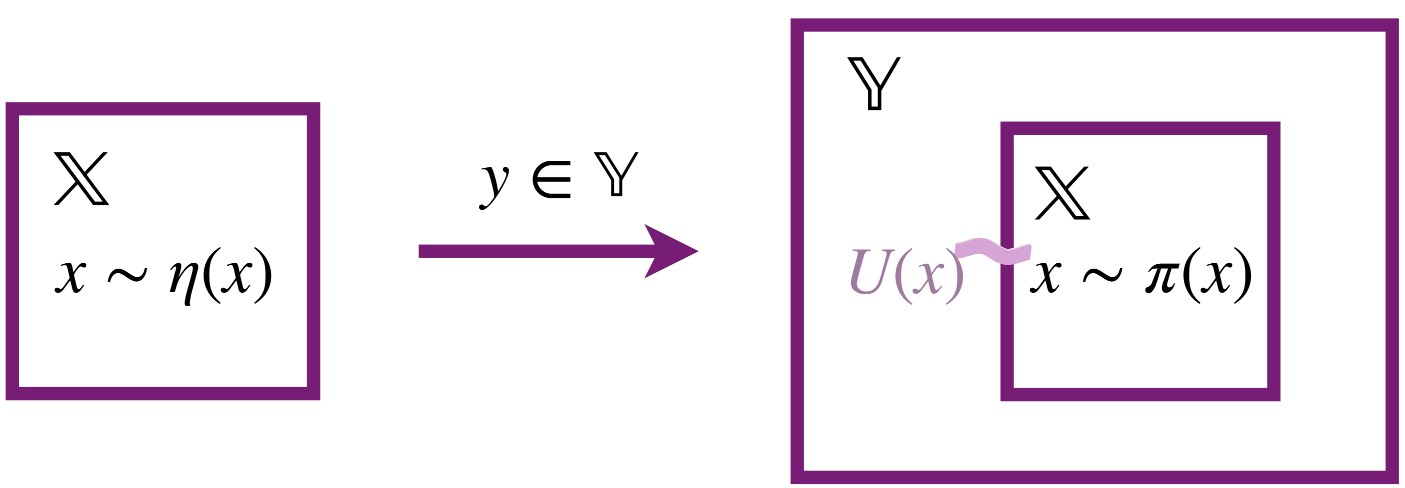

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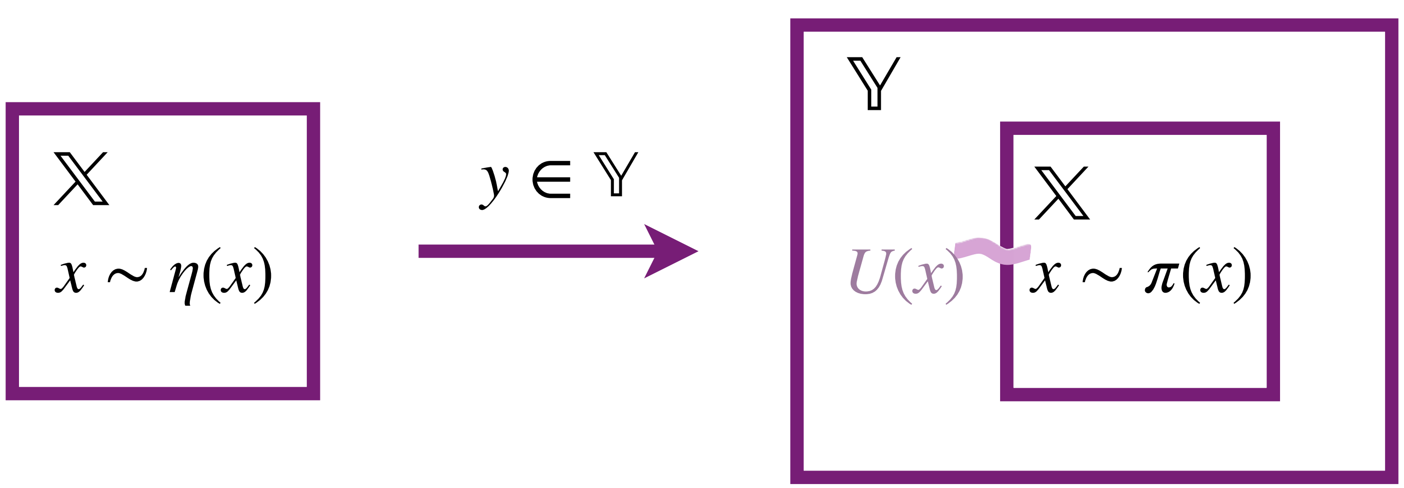

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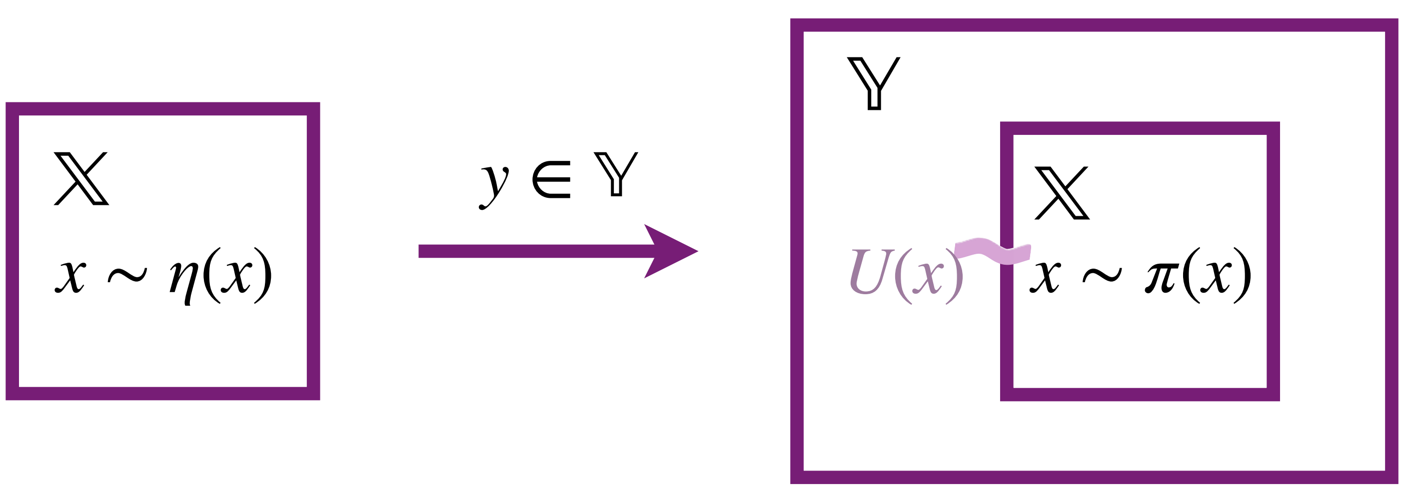

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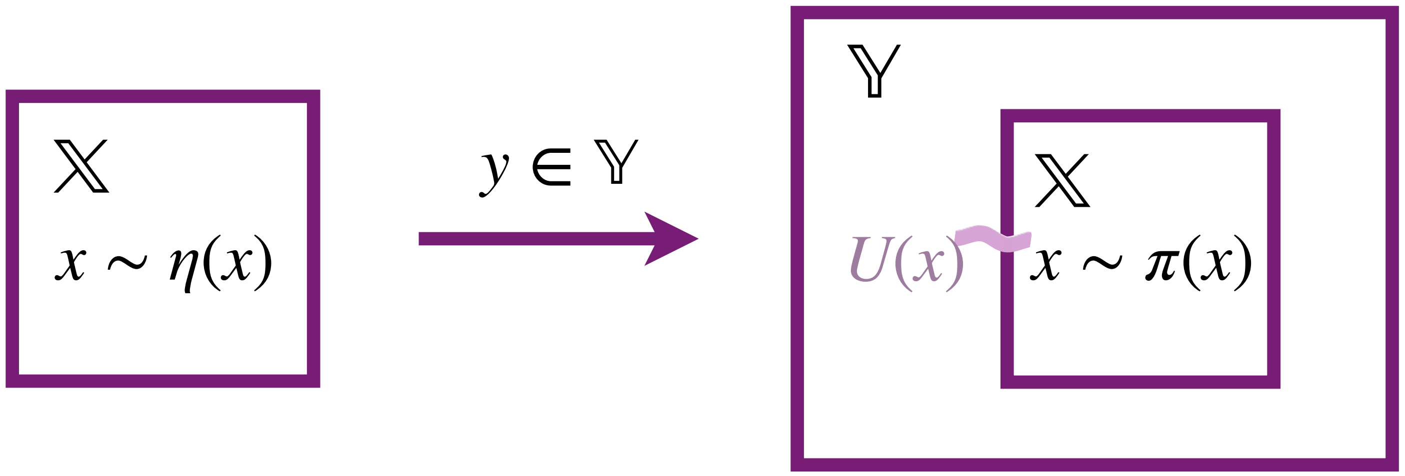

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  - Equivalently, we have $U(x)$ is a sufficient statistic for the reservoir state.
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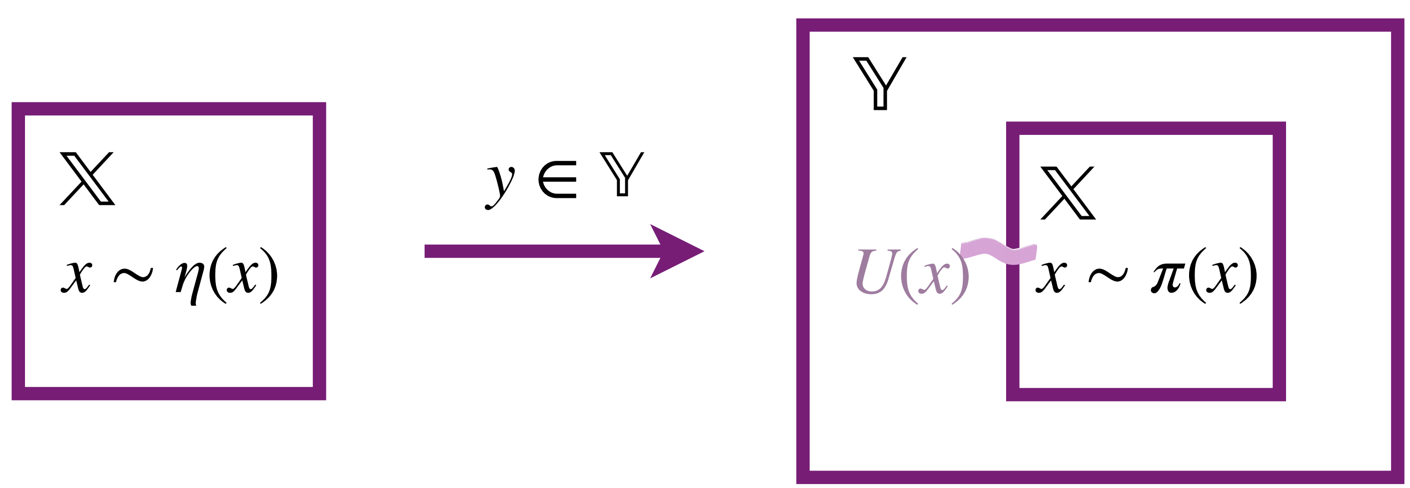
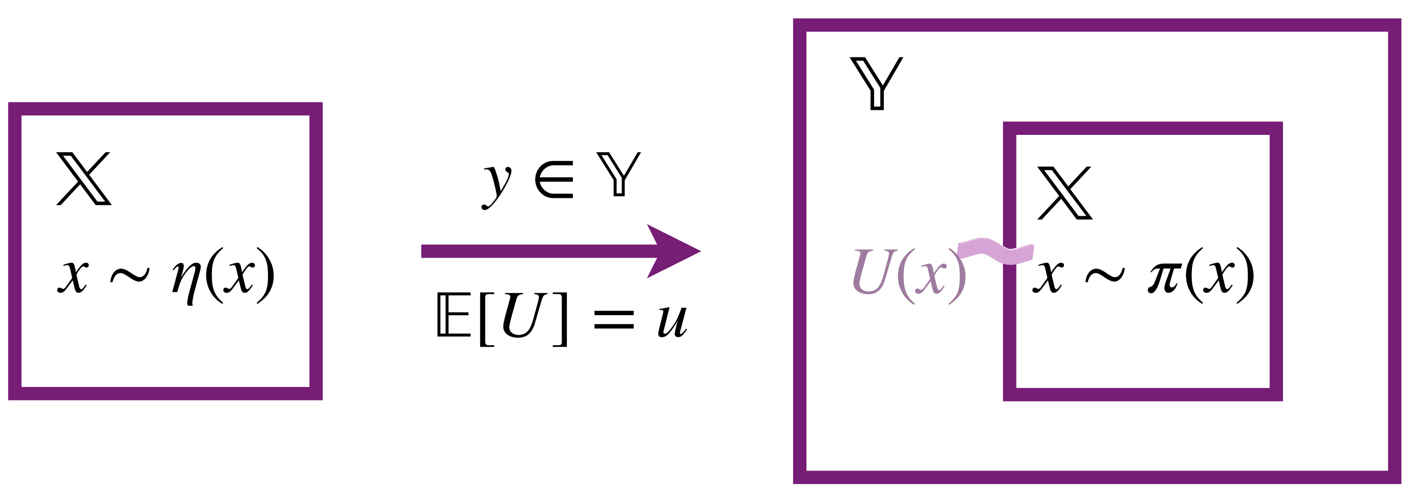
  - By the Neyman-Fisher factorization theorem, for any state $y \in \mathbb{Y}$ we have

$$\pi(\mathrm{d}x) = \eta(\mathrm{d}x) g(U(x), y)$$



POSTULATE 3



$\mathbb{X}$
$x \sim \eta(x)$
$y \in \mathbb{Y}$
$\mathbb{E}[U] = u$
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$U(x)$
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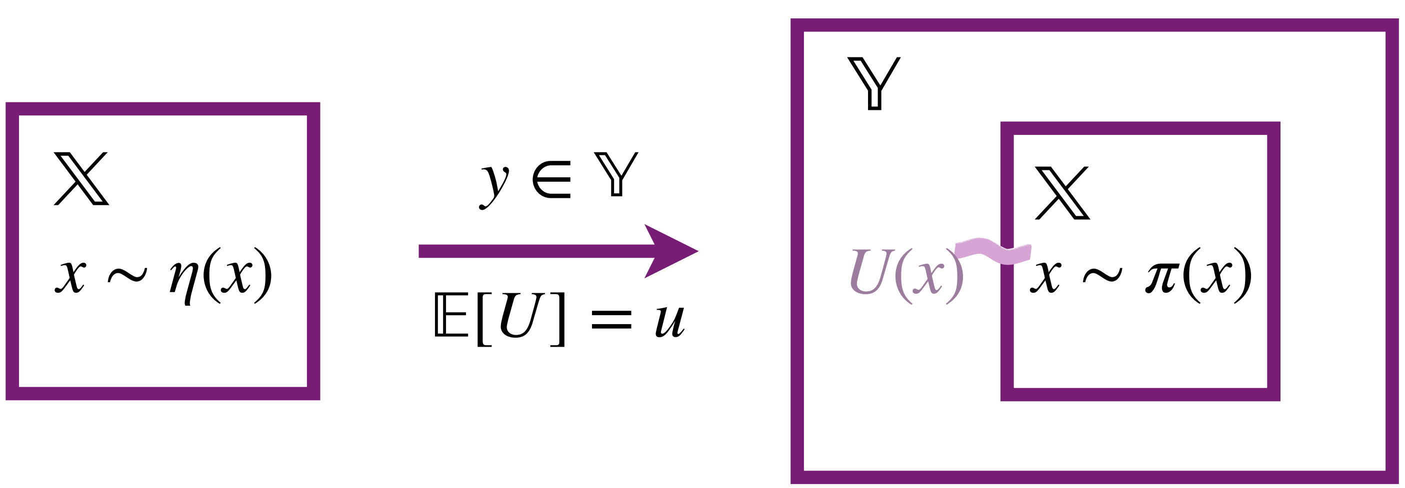
POSTULATE 3

- The system and resevoir state dynamically evolve and will fluctuate even at at stationarity



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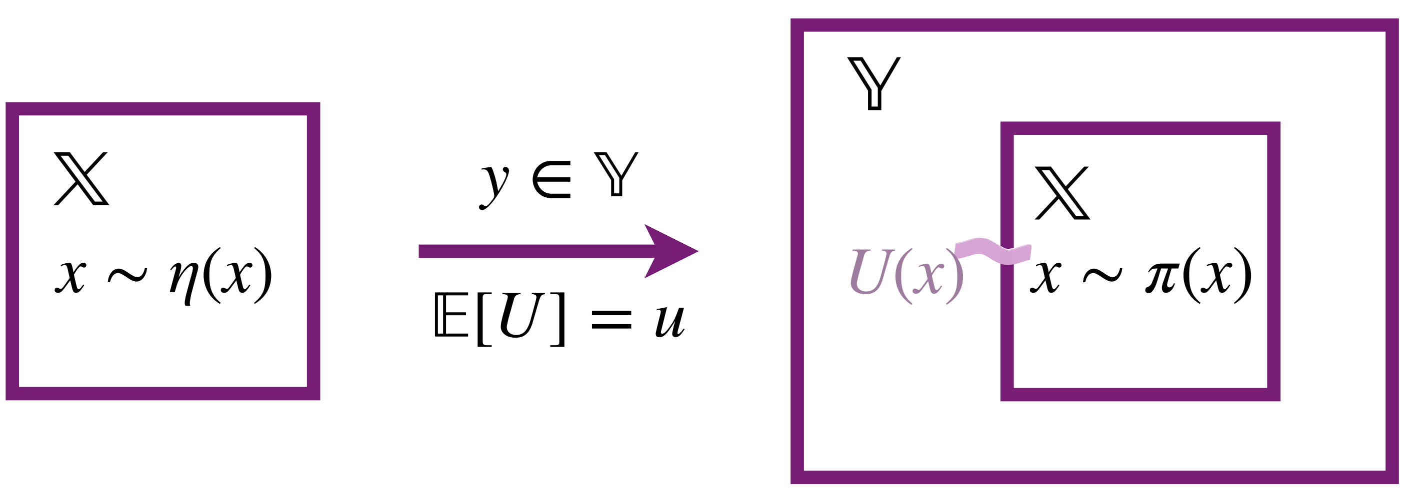

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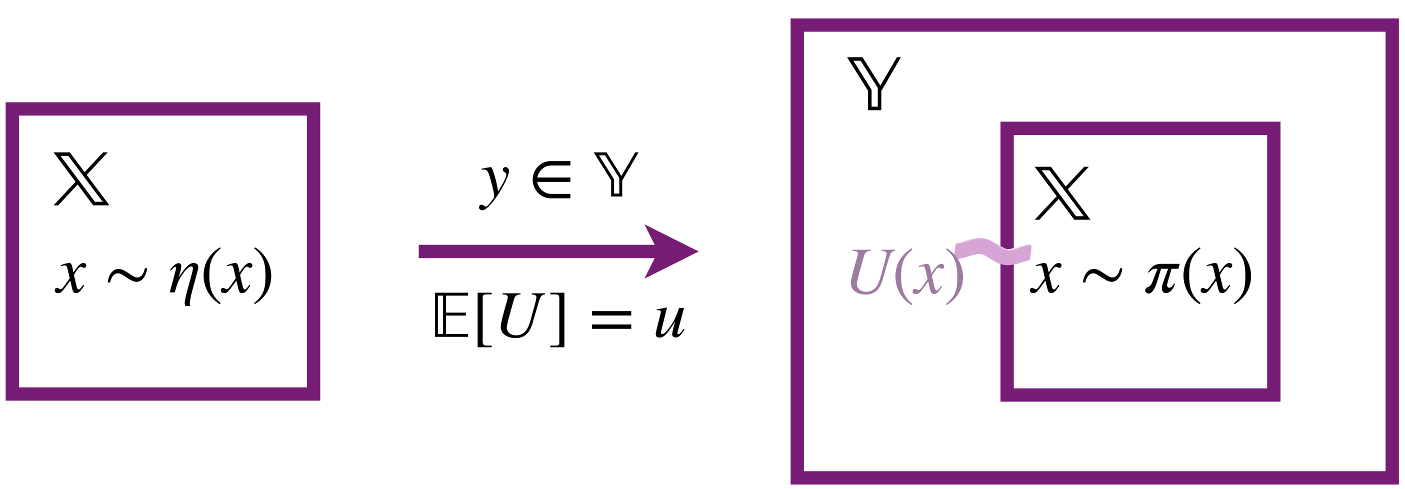

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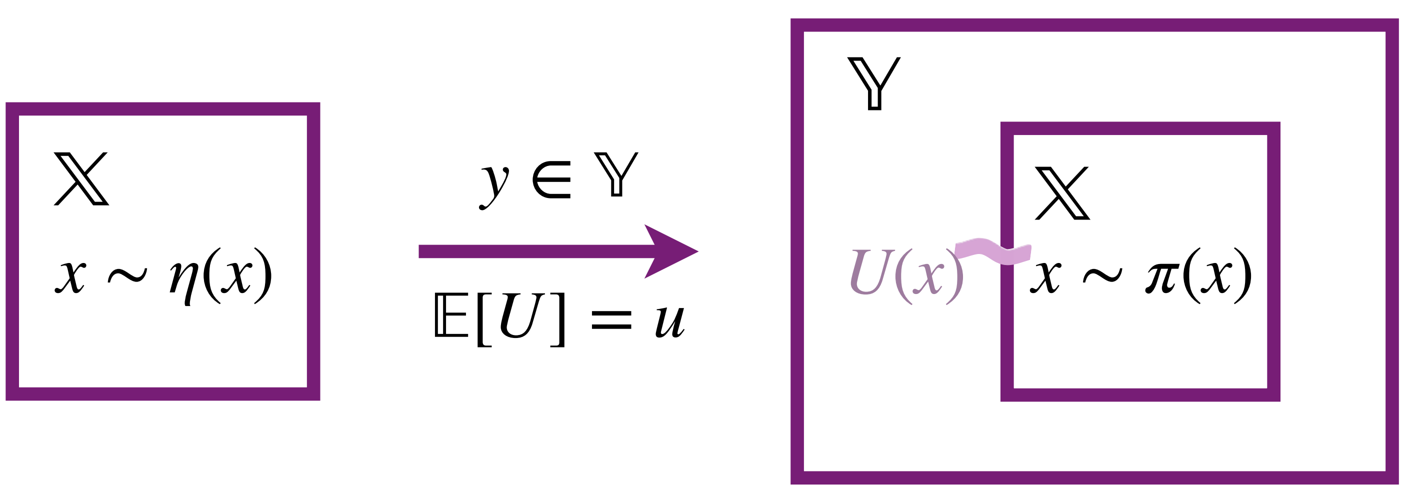
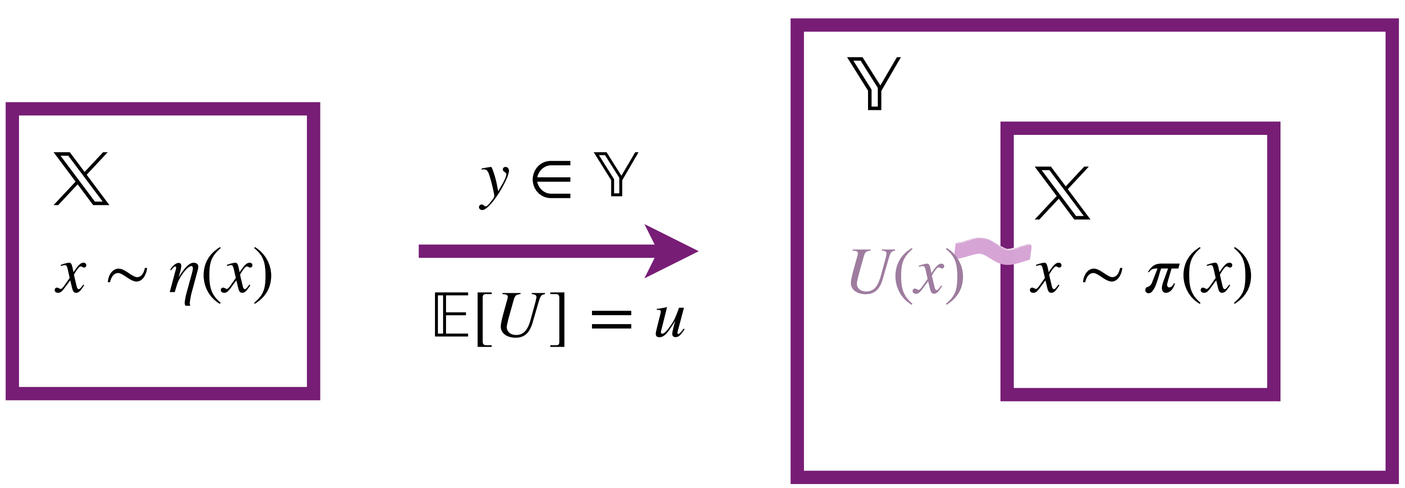
  - All other details of the reservoir are irrelevant. The system knows nothing about the reservoir except this single number ϵ



POSTULATE 4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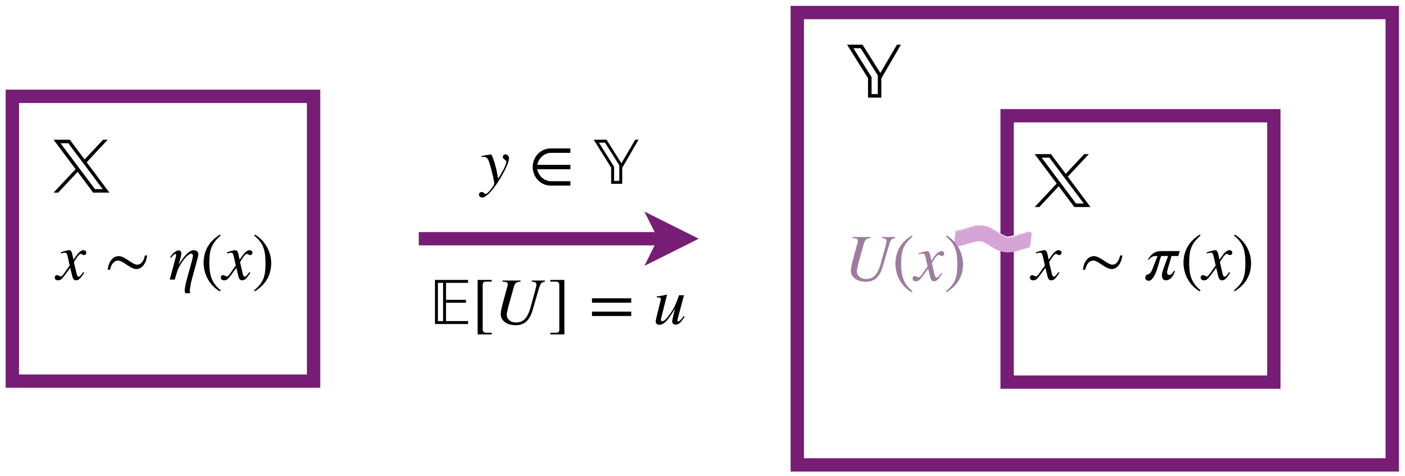

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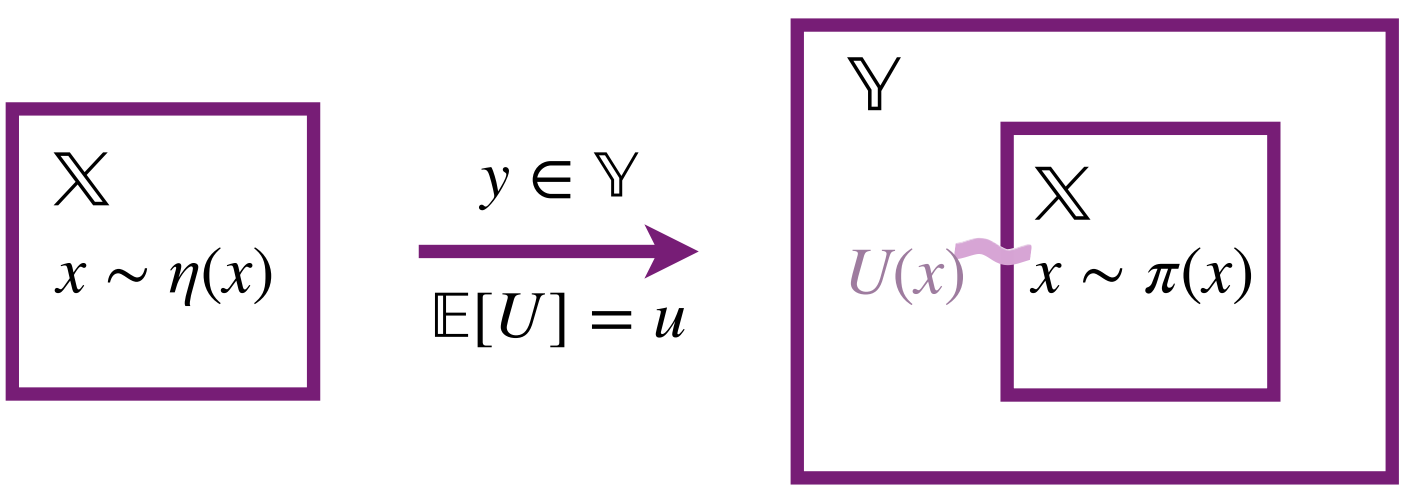
- Closeness is measured by KL divergence:

$$\text{minimize}: \qquad \mathrm{KL}(\pi \| \eta) = \pi\left[\log \frac{\mathrm{d}\pi}{\mathrm{d}\eta}\right]$$

$$\text{Constraint}: \qquad \pi[1] = 1, \qquad \pi[U] = u$$



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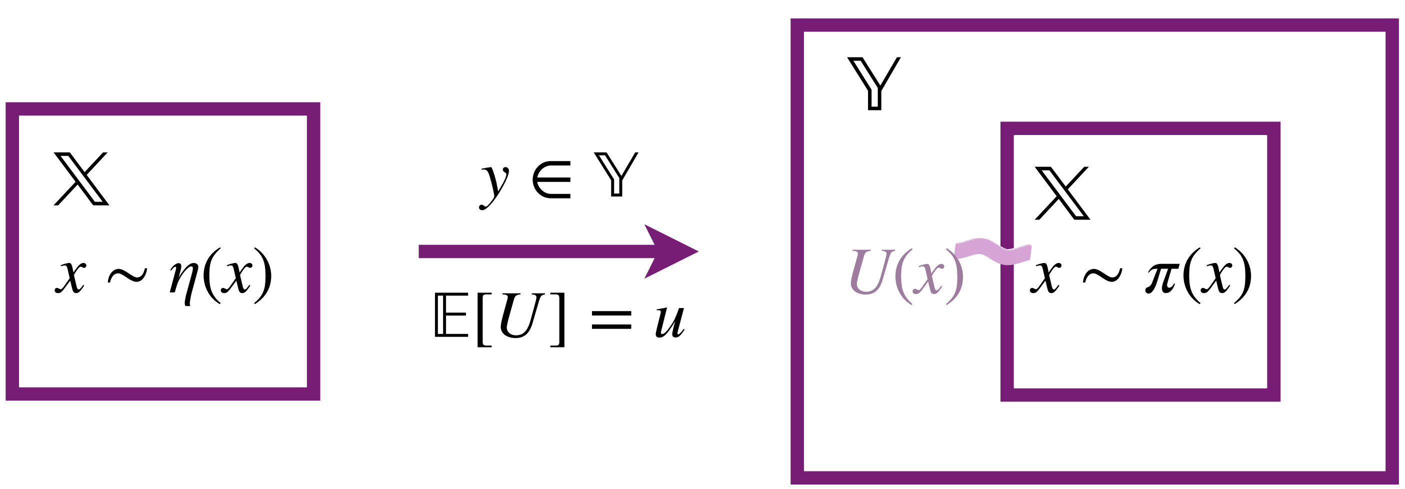

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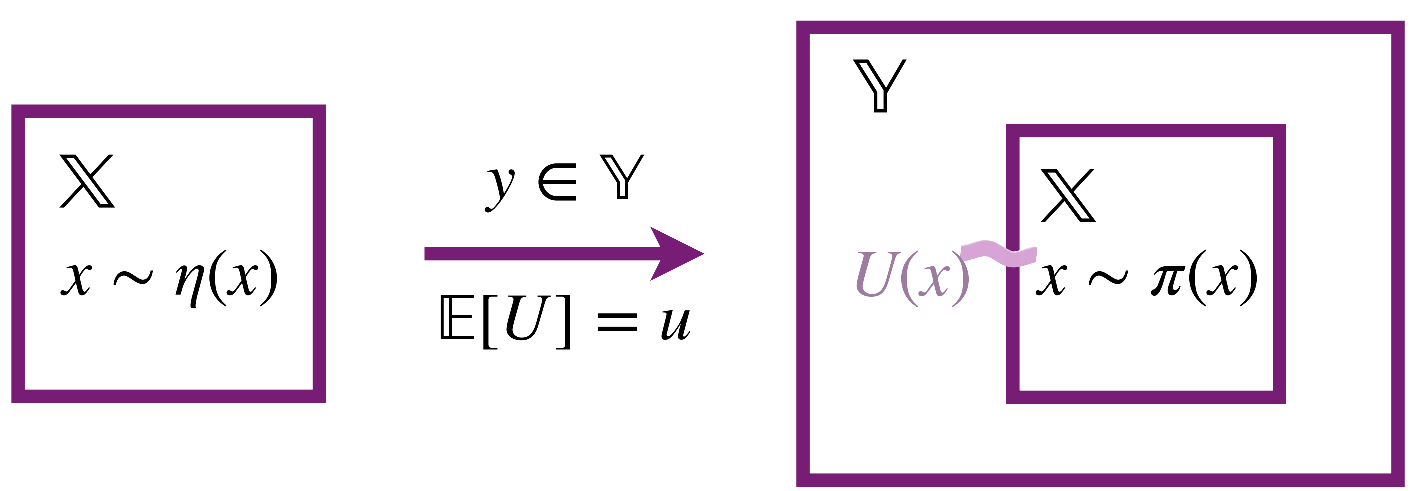
  - In chemistry we introduce the Boltzmann constant $k_B = 1.380649 \times 10^{-23}$ joules per Kelvin



POSTULATE 4

- **Postulate 4:** Given the reference and the energy constraint, the equilibrium distribution π is the one that is **as close to η as possible** while satisfying the constraint.
- Closeness is measured by KL divergence:

$$\text{minimize}: \qquad \mathrm{KL}(\pi \| \eta) = \pi\left[\log \frac{\mathrm{d}\pi}{\mathrm{d}\eta}\right]$$

$$\text{Constraint}: \qquad \pi[1] = 1, \qquad \pi[U] = u$$

- We define the relative entropy of π with respect to η as

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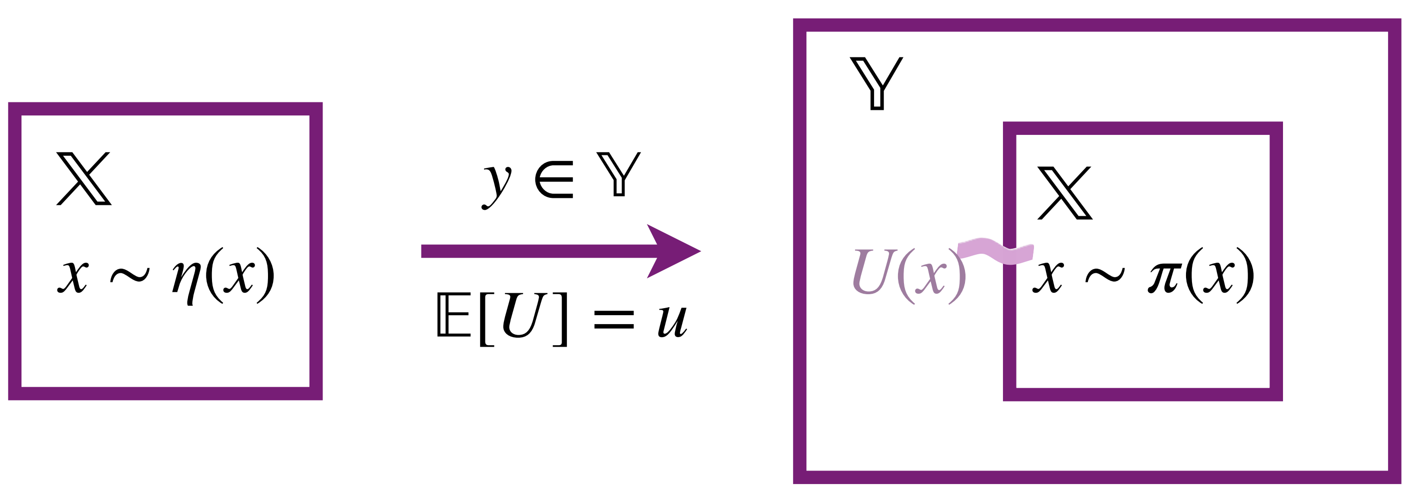
- Postulte 4 is Jaynes' maximum entropy principle: “don’t assume anything beyond the constraint”



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- Again, we can assume $k_B = 1$ but in the physics literature, you will see $\beta^{-1} = k_B T$

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- We see that there is a duality, as β deceases, we can assume the density is flattening

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- From now on we will assume $k_b = 1$

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- Therefore partition function, and entropy encode the same information!

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- A system is in equilibrium if and only if $F = 0$

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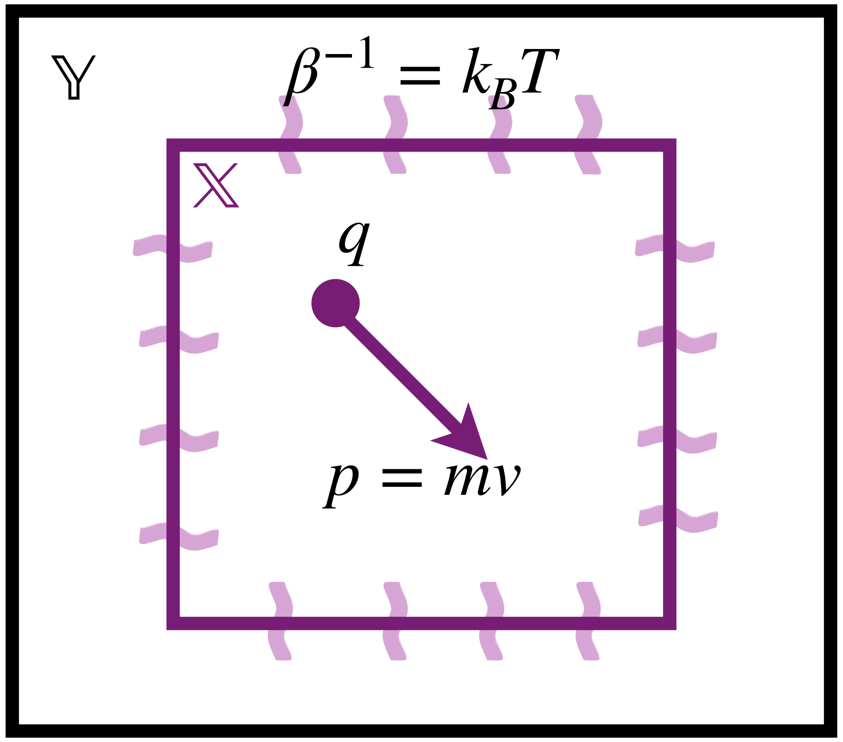

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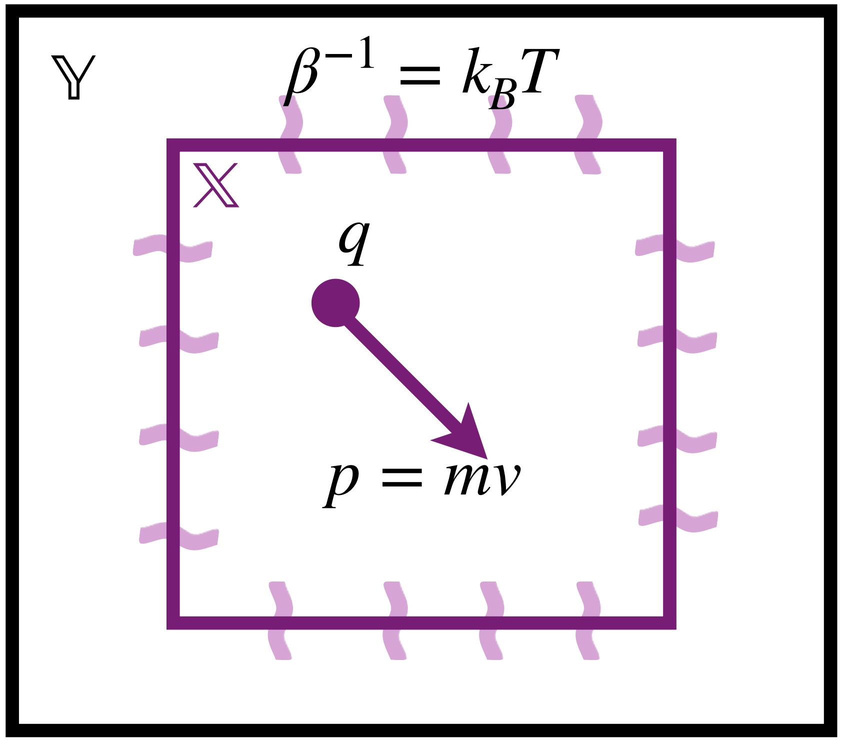

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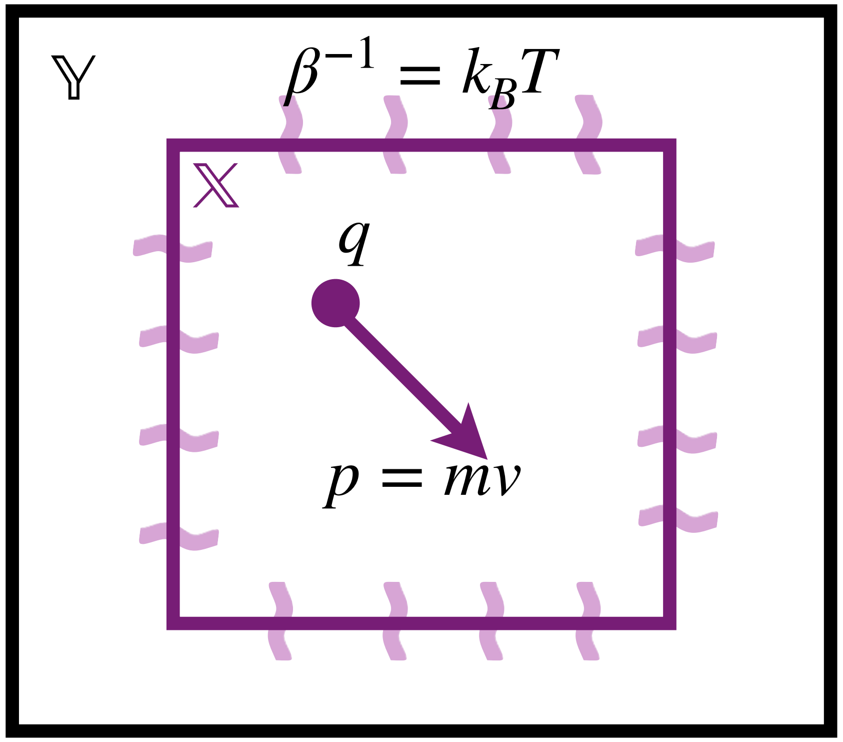

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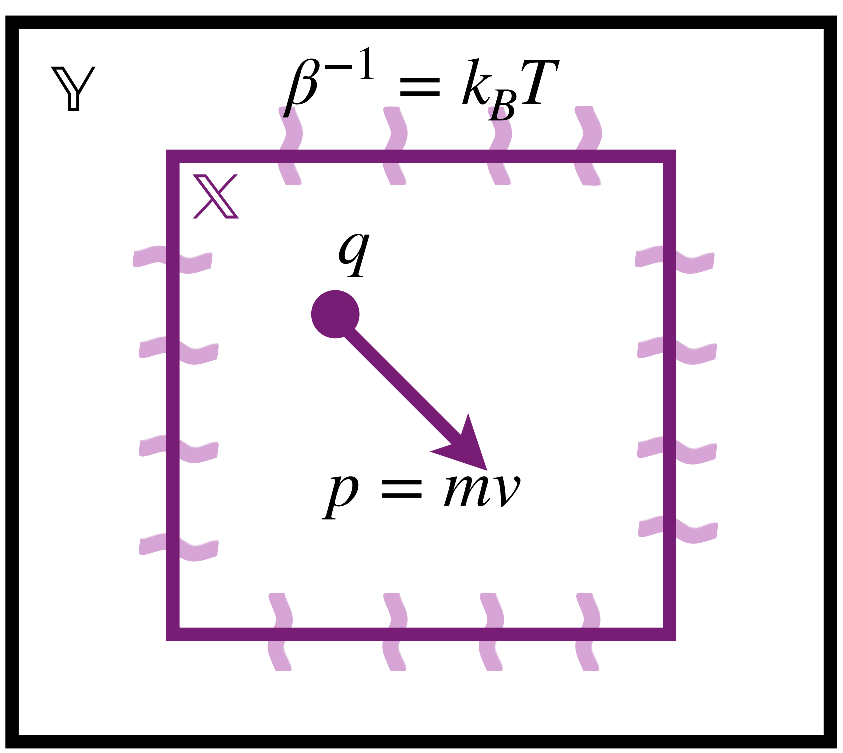

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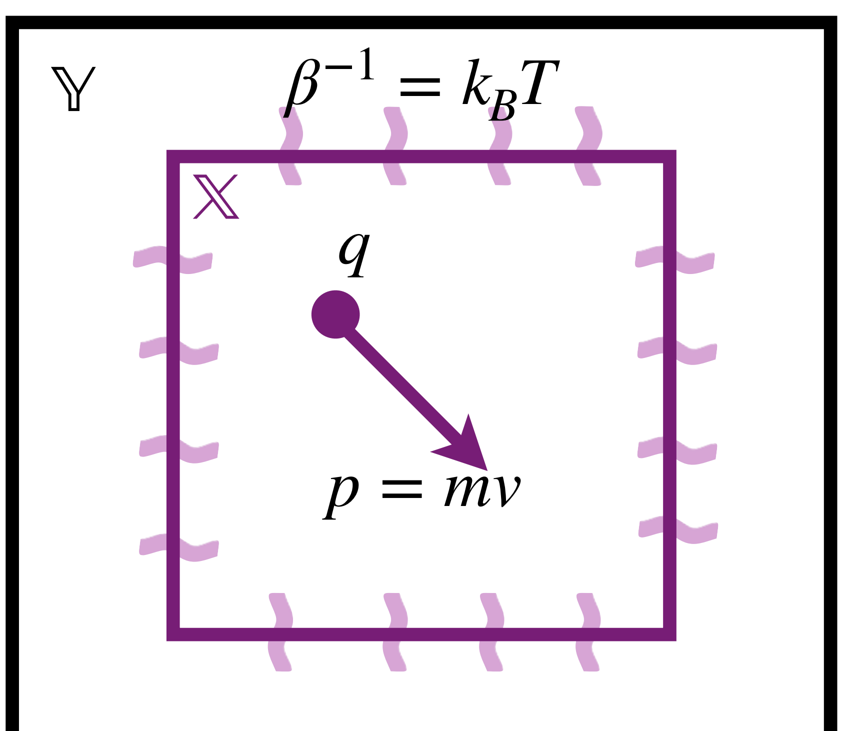

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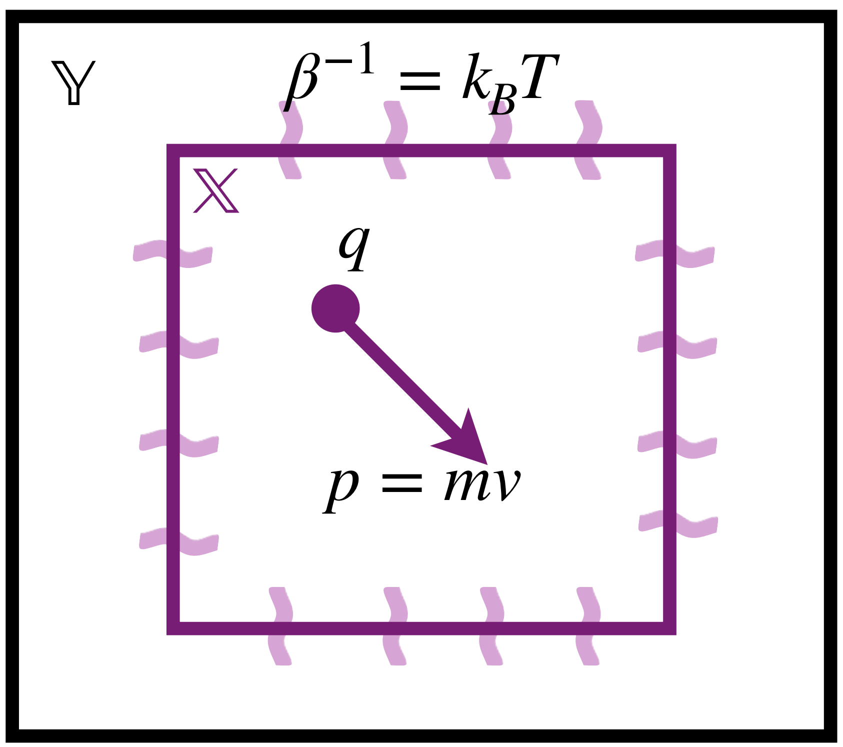

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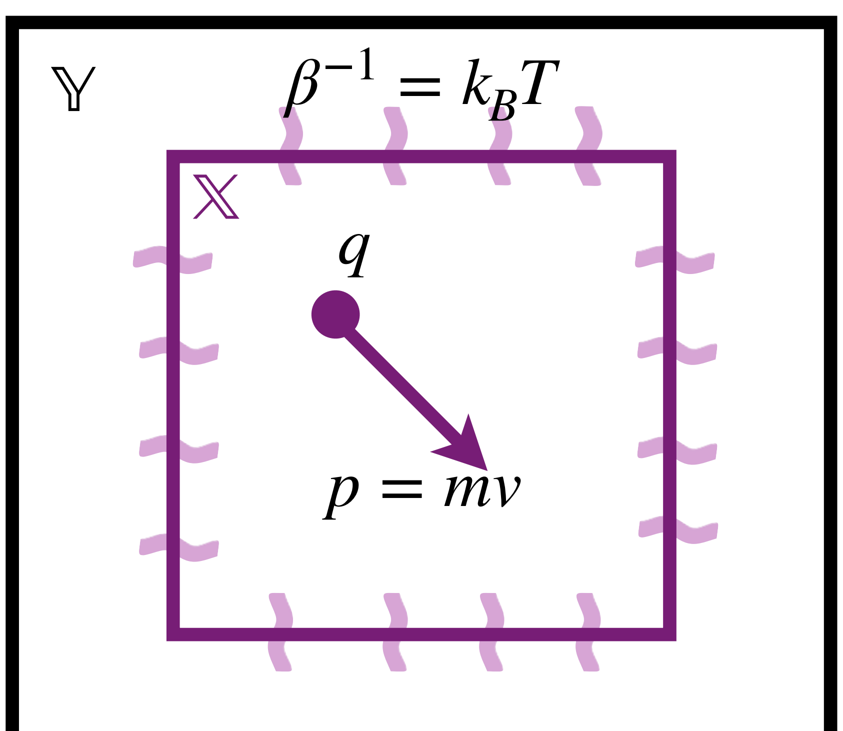

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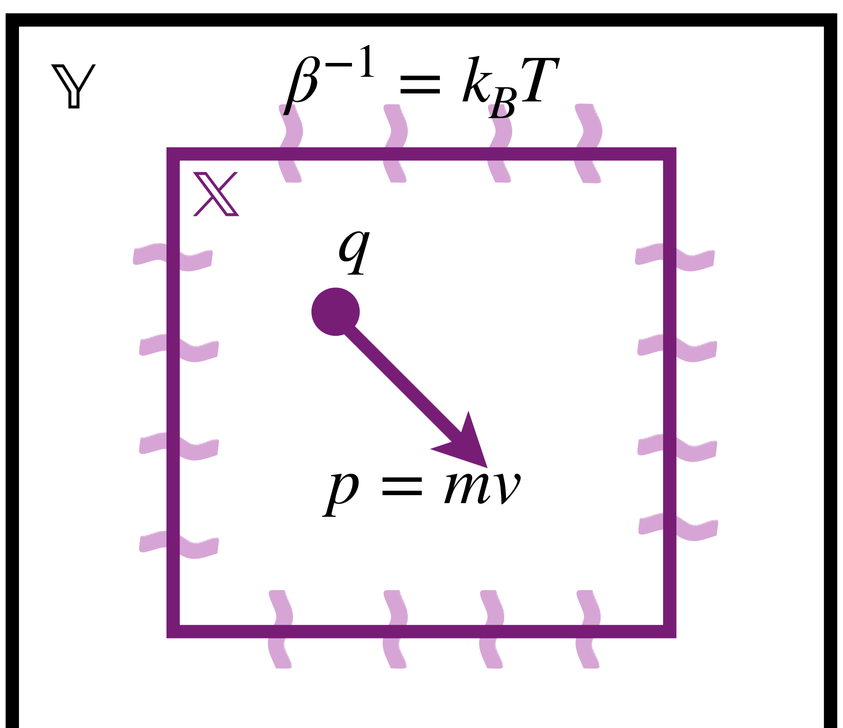

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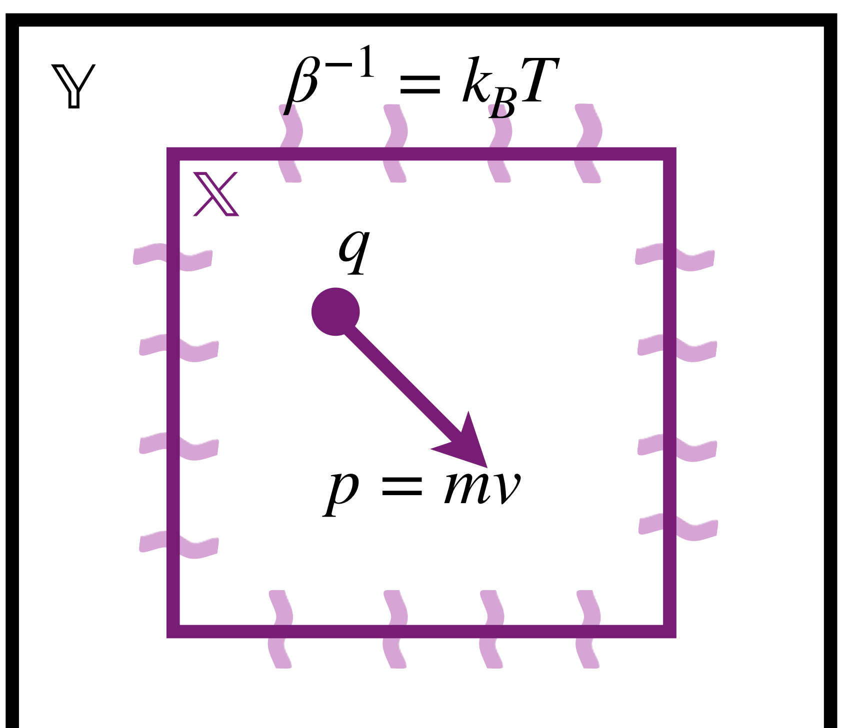

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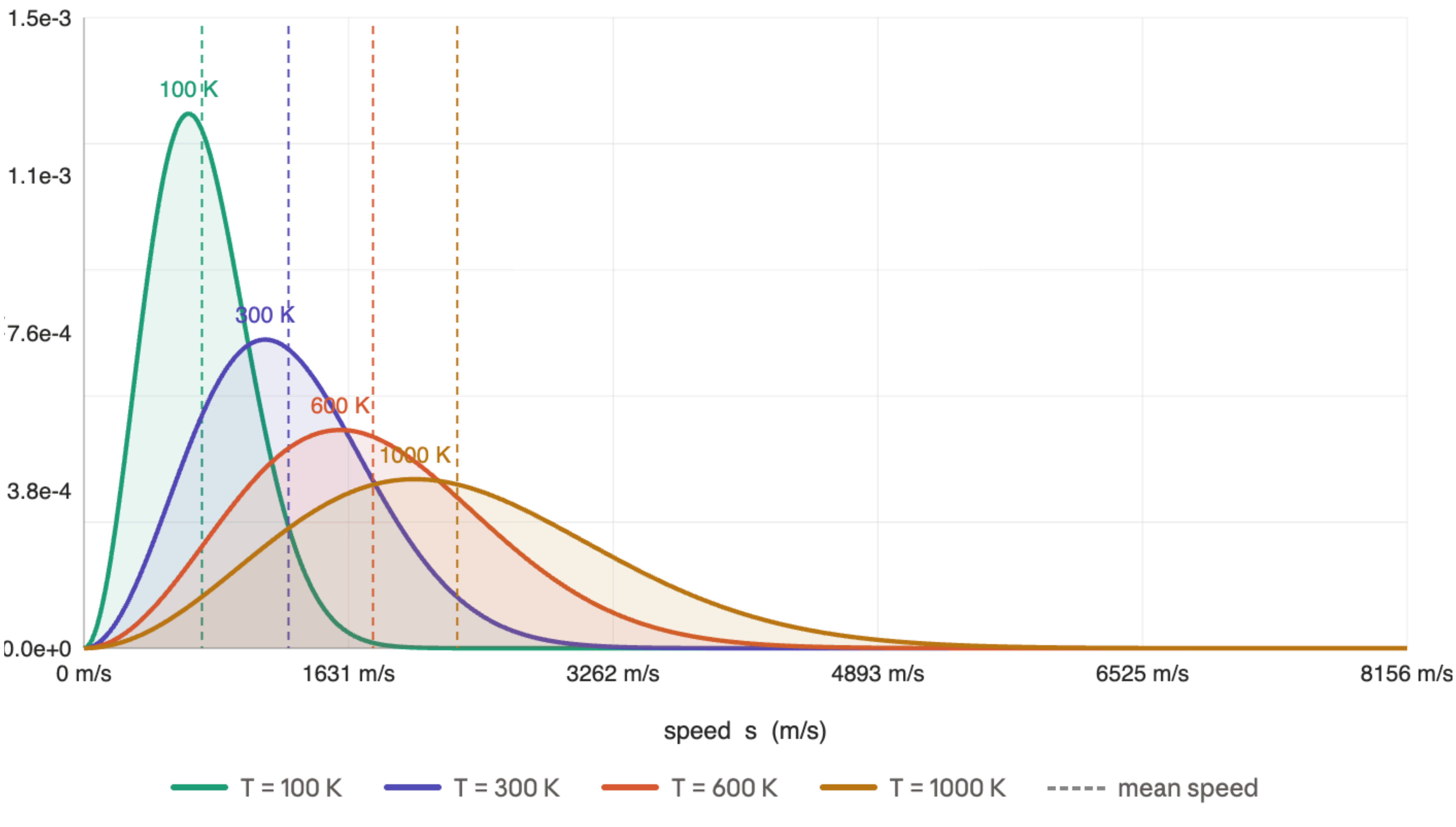

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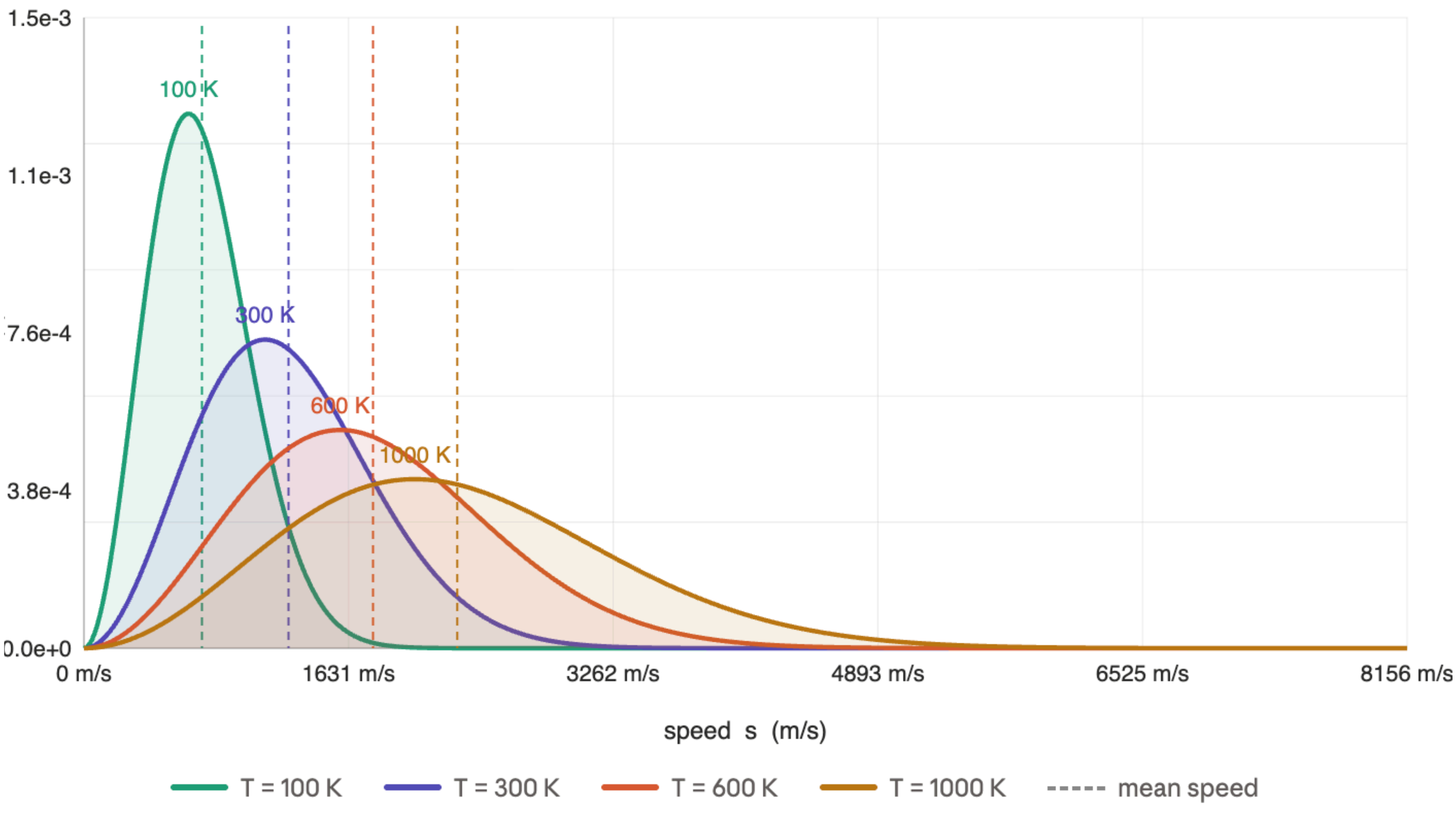
- With $m = 4u$ we have



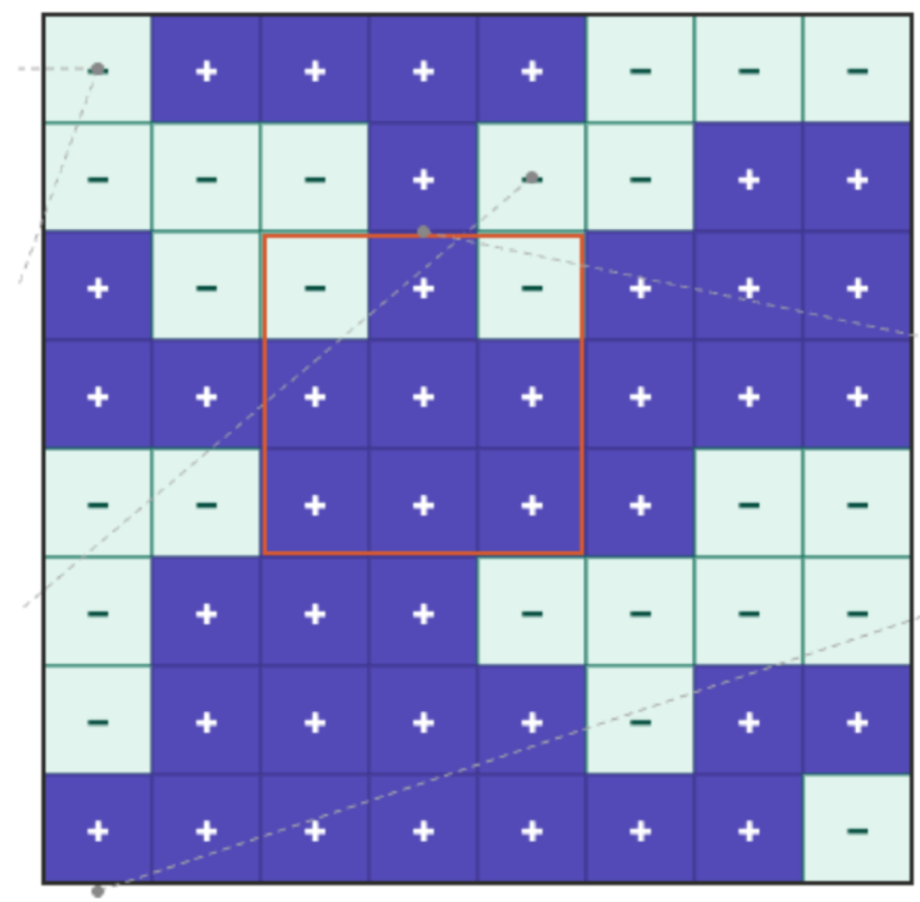
EXAMPLE: ISING MODEL

- Let $\mathbb{X} = \{-1,1\}^V$ where V is the a d-dimensional lattice
- The energy forces neighbouring spins to prefer alignment, penalised by an external field

$$U(x) = -J\sum_{i\sim j} x_i x_j - h\sum_i x_i$$

  - The first term favours aligned neighbours with coupling parameter J
  - The second term favours all spins aligning with magnetic moment h

$$\pi_\beta(x) \propto \exp\left(\beta J\sum_{i\sim j} x_i x_j + \beta h\sum_i x_i\right)$$



  - Total magnetisation of the state x is $m(x)$

$$m(x) = \sum_{i=1} x_i$$

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  - $\beta > \beta_c$ we have $m(x) \approx \pm m_*(\beta)$

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$\beta = 0.15$ $\qquad$ $\beta = 0.44$ $\qquad$ $\beta = 1.5$